MAT 4199, Winter 2018

Assignment 3

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Due date: March 22, 2018

1. **[2 points]** Recall that if (ρ, V) is a $\mathfrak{g}$-module for a Lie algebra $\mathfrak{g}$, then the module (ρ^*, V^*) is defined as follows:

$$\big(\rho^*(x)\phi\big)(v) := -\phi(\rho(x)v) \quad \text{for } x \in \mathfrak{g},\ \phi \in V^*,\ v \in V.$$

Let (ρ, V) be a $\mathfrak{g}$-module. Assume that (ρ, V) and (ρ^*, V^*) are isomorphic $\mathfrak{g}$-modules. Prove that there exists a non-degenerate bilinear form

$$\langle \cdot, \cdot \rangle : V \times V \to \mathbb{C}$$

such that $\langle \rho(x)v, w\rangle + \langle v, \rho(x)w\rangle = 0$ for all $x \in \mathfrak{g}$ and $v, w \in V$.

2. Let $V := \mathbb{C}^n$ be the standard $\mathfrak{gl}_n$-module, i.e., any $x \in \mathfrak{gl}_n$ acts on any $v \in \mathbb{C}^n$ by matrix multiplication on the left.

(a) **[2 points]** Is it true that $V \cong V^*$ as $\mathfrak{gl}_n$-modules?

(b) **[2 points]** Determine the highest weight of V with respect to the Borel subalgebra $\overline{\mathfrak{b}}$, consisting of *lower* triangular $n \times n$ matrices.

3. Let $V := \mathrm{Sym}_{n\times n}(\mathbb{C})$, the space of $n \times n$ symmetric matrices. Consider the following map:

$$\rho : \mathfrak{gl}_n \to \mathrm{End}_{\mathbb{C}}(V)\ ,\ \rho(X)A := XA + AX^T.$$

(a) **[2 points]** Prove that (ρ, V) is indeed a $\mathfrak{gl}_n$-module.

(b) **[2 points]** Compute the weights of V with respect to the standard Cartan subalgebra of $\mathfrak{gl}_n$.

(c) **[2 points]** Using (b) prove that V is an irreducible $\mathfrak{gl}_n$-module.

Hint: Show that by the action of $\mathfrak{gl}_n$ you can "reach" from any weight space to any other weight space.